



An Entropy-Weighted Group Decision-Support Framework for Prioritizing Antihypertensive Drug Classes in Older Patients With Comorbidities

Hamed Fazlollahtabar^{1*}, Zahra Hassanzadeh²

¹ Department of Industrial Engineering, School of Engineering, Damghan University, 36716-45667 Damghan, Iran

² Department of Industrial Engineering, Mazandaran University of Science and Technology, 47166-85635 Babol, Iran

* Correspondence: Hamed Fazlollahtabar (h.fazl@du.ac.ir)

Received: 07-24-2026

Revised: 09-14-2026

Accepted: 09-20-2026

Citation: H. Fazlollahtabar and Z. Hassanzadeh, “An entropy-weighted group decision-support framework for prioritizing antihypertensive drug classes in older patients with comorbidities,” *J. Intell Manag. Decis.*, vol. 5, no. 3, pp. 235–255, 2026. <https://doi.org/10.56578/jimd050304>.



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Abstract: Selecting an appropriate antihypertensive drug class for older patients with multimorbidity requires multiple clinical considerations to be evaluated simultaneously, including comorbidity-specific suitability, treatment-related risks, therapeutic priorities and professional judgement. A transparent decision-support framework is therefore needed to structure these heterogeneous considerations without implying that a mathematical ranking constitutes a clinical recommendation. An entropy-weighted group decision-support framework was developed and evaluated using a hypothetical 72-year-old patient with multiple comorbidities. Seven antihypertensive drug classes—diuretics, beta-blockers, Angiotensin-Converting Enzyme (ACE) inhibitors, Angiotensin II Receptor Blockers (ARBs), calcium-channel blockers (CCBs), alpha-1 blockers and central alpha-2 agonists—were assessed against eight criteria: physician experience, suitability for older patients, suitability for patients with diabetes, suitability for patients with kidney disease, suitability for patients with congestive heart failure, suitability for patients with a history of myocardial infarction, medication-related complication risk and rapidity of therapeutic effect. Assessments were provided independently by an internist, a cardiologist and a urologist. Criterion weights were derived using the entropy method from transformed rank-score distributions, while rank-frequency linear assignment and the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) were used to aggregate expert assessments and obtain alternative rankings. ARBs were ranked first by TOPSIS and third in both optimal linear-assignment solutions. Two equally optimal linear-assignment solutions were obtained, with beta-blockers and alpha-1 blockers exchanging the first and sixth positions. Central alpha-2 agonists were ranked last by both approaches. Importantly, the complete and tie-free rankings provided by every expert resulted mathematically in identical entropy weights of 0.125 for all eight criteria, indicating that criterion differentiation was not achieved under the adopted elicitation format. The resulting rankings therefore represent methodological outputs rather than evidence of clinical superiority among antihypertensive drug classes. The framework provides a transparent means of structuring multi-criteria and multi-expert assessments in complex clinical scenarios, while its preliminary nature, limited expert panel and absence of patient-level validation preclude direct clinical application. Validation using larger and more diverse expert panels, clinically validated criteria and patient-level outcomes is warranted before the framework can be considered for clinical decision-support applications.

Keywords: Clinical decision support; Group decision-making; Antihypertensive drug prioritisation; Entropy weighting; TOPSIS; Rank-frequency linear assignment

1 Introduction

Arterial hypertension continues to represent one of the leading modifiable risk factors for global cardiovascular disease, stroke, and premature mortality. The World Health Organization estimated that approximately 1.4 billion adults worldwide aged 30–79 years had hypertension in 2024, with about two-thirds residing in low- and middle-income countries [1]. Despite advances in antihypertensive therapy and widespread implementation of clinical practice guidelines, achieving and maintaining adequate blood pressure control remains challenging for many

patients with hypertension. Approximately 23% of adults with hypertension in this age range had their condition under control [1].

A critical barrier to optimal hypertension control is the well-documented phenomenon of clinical inertia—defined as the failure of healthcare providers to initiate or intensify therapy when treatment targets are not achieved [2, 3]. Clinical inertia represents a significant public health concern because uncontrolled hypertension directly contributes to preventable cardiovascular events, stroke, kidney failure, and premature death. Patients with uncontrolled hypertension frequently present in an asymptomatic state, which paradoxically reduces the perceived urgency for therapeutic intervention among both clinicians and patients. This inertia is not monolithic; rather, it is shaped by a combination of interacting patient-specific, physician-related, and system-level factors.

The multifactorial drivers of clinical inertia include competing clinical priorities and the simultaneous presence of comorbid conditions such as type 2 diabetes mellitus, dyslipidemia, and cardiovascular disease [4–7]. Specifically, longitudinal clinical observations demonstrate that therapeutic modification is frequently delayed in patients with poorly controlled hypertension, dyslipidemia, and diabetes despite persistently elevated blood pressure above recommended targets [5].

To address complex clinical decision-making in multi-stakeholder healthcare environments, operations research (OR) methodologies offer structured quantitative frameworks [8, 9]. These analytical tools enable systematic evaluation of trade-offs among competing clinical criteria without relying solely on ad hoc heuristics [10, 11]. For example, the presence of diabetes may favour the use of renin-angiotensin system blockers due to their renoprotective effects, while prior myocardial infarction may compel the use of beta-blockers for their established mortality benefit.

Documentation gaps and suboptimal therapeutic adjustment remain common in inpatient care and chronic disease management. For instance, retrospective studies of inpatient diabetes care demonstrate that hyperglycemia is often under-documented and under-treated during hospitalization, reflecting pervasive clinical inertia in acute care settings [12].

Therapeutic inertia is frequently observed in inpatient glycaemic management, where elevated blood glucose levels and persistent hyperglycemia fail to prompt timely treatment intensification during hospital stays [12, 13]. Such recurring patterns of clinical inertia across acute care settings highlight the need for systematic, evidence-based decision frameworks [13].

To streamline treatment selection and reduce clinical variability, multi-criteria decision-making (MCDM) and structured decision-support applications have been increasingly deployed [10, 11]. When appropriately designed and validated, these decision-support tools assist clinicians in balancing competing therapeutic priorities, improving decision quality while preserving clinical autonomy and patient safety [11].

Selecting antihypertensive therapy for older patients with multimorbidity requires consideration of age, coexisting disease, target-organ damage, concomitant medications, adverse effects, and the need for a timely therapeutic effect. Polypharmacy and competing clinical indications can make these decisions difficult to implement consistently in routine care. Current hypertension guidelines provide a basis for individualized risk-benefit assessment, including consideration of overtreatment and orthostatic hypotension in frail older patients [14–16]. Seven drug classes are considered in this study; their mechanisms of action, clinical indications, and tolerability profiles are outlined below.

Diuretics: Thiazide and thiazide-like agents (hydrochlorothiazide, chlorthalidone), potassium-sparing agents such as triamterene, and loop diuretics (furosemide, bumetanide) primarily reduce extracellular fluid volume and vascular resistance. Thiazides are preferred for first-line therapy due to their proven efficacy in reducing cardiovascular events. Loop diuretics have particular roles in managing volume overload in heart failure and substantial renal impairment. The 2025 American Heart Association (AHA)/American College of Cardiology (ACC), 2023 European Society of Hypertension (ESH), and 2024 European Society of Cardiology (ESC) guidelines all recommend thiazide diuretics as first-line agents for most patients with hypertension, particularly those with salt-sensitive hypertension or who are at risk of heart failure. Potential metabolic adverse effects of thiazide therapy, including hypokalemia, hyperglycemia and dyslipidemia, should be considered alongside its established benefits [14–16].

Beta-blockers: Atenolol, propranolol, metoprolol, bisoprolol, and carvedilol competitively antagonize beta-adrenergic receptors, reducing cardiac output and renin release. According to the 2025 AHA/ACC guidelines [16], beta-blockers are not recommended as first-line therapy for uncomplicated hypertension. In contrast, the 2023 ESH guidelines [14] retain beta-blockers among the major drug classes suitable for starting and continuing treatment, particularly when compelling indications such as post-myocardial infarction, heart failure with reduced ejection fraction, or angina are present. The evidence for beta-blockers in older patients with heart failure is particularly strong, with multiple trials demonstrating reductions in mortality and hospitalizations.

Angiotensin-converting enzyme (ACE) inhibitors: Captopril, enalapril, ramipril, lisinopril, and perindopril attenuate the renin-angiotensin-aldosterone cascade by inhibiting the conversion of angiotensin I to angiotensin II. ACE inhibitors are recommended as first-line therapy for most patients, with specific indications including albuminuric chronic kidney disease and heart failure with reduced ejection fraction. The evidence supporting ACE inhibitors in these populations is extensive, with multiple large-scale trials demonstrating reductions in cardiovascular

events, progression of kidney disease, and mortality. However, ACE inhibitors can cause a persistent dry cough, which may lead to treatment discontinuation [14–16].

Angiotensin II receptor blockers (ARBs): Losartan, valsartan, candesartan, irbesartan, telmisartan, and olmesartan block angiotensin II type 1 receptors are recommended as first-line alternatives to ACE inhibitors for appropriate patients. They have a lower incidence of cough than ACE inhibitors. The 2023 ESH guidelines and 2025 AHA/ACC guidelines both recommend ARBs as first-line agents, particularly for patients who cannot tolerate ACE inhibitors [14, 16].

Calcium channel blockers (CCBs): Dihydropyridines (amlodipine, nifedipine, felodipine) and non-dihydropyridines (verapamil, diltiazem) inhibit voltage-gated calcium channels. Dihydropyridines act mainly through arterial vasodilation, whereas non-dihydropyridines also reduce cardiac contractility and atrioventricular conduction. Dihydropyridine CCBs are recommended as first-line therapy, particularly for older patients with isolated systolic hypertension. Non-dihydropyridine CCBs should not be used in patients with heart failure with reduced ejection fraction because of their negative inotropic effects [16]. The evidence for CCBs in older patients is robust, with trials demonstrating significant reductions in stroke and cardiovascular events [17].

Alpha-1 blockers: Doxazosin, prazosin, and terazosin block peripheral alpha-adrenergic receptors, causing vasodilation and reducing peripheral vascular resistance. These agents are not recommended as first-line therapy for hypertension due to less favourable cardiovascular outcomes in clinical trials (notably the ALLHAT trial) and a substantial risk of orthostatic hypotension. However, they may be considered for patients with concomitant benign prostatic hyperplasia (BPH), as they provide symptomatic relief for lower urinary tract symptoms. This dual benefit is particularly relevant in older male patients.

Central alpha-2 agonists: Methyldopa and clonidine reduce sympathetic outflow from the central nervous system by stimulating alpha-2-adrenergic receptors in the brainstem. These agents are generally reserved for patients with resistant hypertension or for specific clinical circumstances (e.g., methyldopa in pregnancy) due to side-effect profiles including sedation, dry mouth, dizziness, and rebound hypertension upon abrupt discontinuation. Their use has declined substantially with the availability of better-tolerated agents.

Sodium nitroprusside and organic nitrates were excluded from the primary ranking because the analysis focuses on chronic oral blood pressure management rather than acute-care treatment.

Alongside these pharmacological considerations, healthcare OR offers analytical tools to address complex resource allocation and treatment selection problems [9, 18]. OR methodologies applied to healthcare range from discrete-event simulation and mathematical programming to multi-criteria decision analysis [10, 19]. These approaches are particularly advantageous in environments characterized by multiple stakeholders, conflicting objectives, and substantial clinical heterogeneity [11, 20].

Clinical inertia and treatment gaps are further compounded by demographic and sex disparities in routine risk-factor management [21–23]. For instance, empirical evidence demonstrates significant sex differences in dyslipidemia testing and lipid-lowering treatment intensity among managed-care patients with type 2 diabetes, leading to unaddressed cardiovascular risk [24].

Collaborative efforts between academic researchers and clinical practitioners have catalyzed the development of decision-support applications across healthcare domains [10, 11, 25]. These applications demonstrate the potential of structured analytical approaches to organize clinical priorities and improve decision quality while maintaining patient safety and respecting clinical autonomy.

Kidney transplant recipients represent a particularly vulnerable subgroup in whom stringent blood pressure management requires careful consideration of both allograft function and cardiovascular risk [26].

The complexity of these decisions, together with the volume of relevant clinical information, creates a compelling need for structured decision-support tools that can organize and present information in a clinically useful format. Such tools should complement rather than replace clinical judgement, providing a transparent and systematic framework for evaluating trade-offs between different therapeutic options.

The present investigation directly addresses this gap by proposing a hybrid group decision-making framework that combines formula-derived criterion weighting with subjective expert input to rank seven antihypertensive drug classes within a specified hypothetical clinical scenario. The framework is designed to be transparent, replicable, and adaptable to different clinical contexts. By explicitly incorporating multiple criteria and expert perspectives, it seeks to make the reasoning behind drug prioritization more explicit and defensible.

The specific objectives of this study are to: (i) identify and operationalize the key clinical determinants of antihypertensive prescribing in elderly patients with comorbidities, (ii) apply two established group MCDM techniques—rank-frequency linear assignment and the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS)—to generate and compare drug-class rankings, (iii) compare the outputs of these two methods to understand how different aggregation approaches affect final recommendations, and (iv) demonstrate the feasibility of the proposed framework using a representative clinical case study.

2 Methods

Multiple-criteria decision analysis (MCDA), often used synonymously with MCDM, constitutes a well-established subdiscipline within OR that provides a suite of mathematical and computational techniques for systematically evaluating a finite set of alternatives against a finite set of performance metrics [10, 27, 28]. MCDM methods draw upon interdisciplinary knowledge spanning mathematics, behavioral decision theory, economic utility theory, computer science, and information systems. Since their inception in the 1960s, MCDM methodologies have generated an extensive corpus of theoretical developments and practical applications across diverse domains [20]. Within the healthcare sector, MCDA has been established as a formal quantitative framework in a range of operational and clinical contexts, including health technology appraisal, formulary prioritization, clinical guideline development, and shared decision-making [11].

In the present investigation, two complementary MCDM approaches—rank-frequency linear assignment and the TOPSIS algorithm—were adopted to rank the seven antihypertensive drug classes based on eight clinically relevant criteria. Criterion weights were calculated using the entropy method, and individual decision-maker inputs were aggregated to derive group consensus rankings. The use of two distinct methods enables comparison of the rankings and provides insight into how different aggregation approaches may influence final recommendations.

2.1 Study Design and Expert Panel

This study employed a decision-modelling approach based on a representative clinical scenario rather than an actual patient cohort. This scenario was chosen to demonstrate the feasibility of the framework while avoiding the ethical and practical challenges of patient data collection. The representative scenario described a patient aged 72 with hypertension (BP 152/88 mmHg), type 2 diabetes (HbA1c 7.2%), prior myocardial infarction (3 years ago), and mild congestive heart failure (NYHA Class II)—a common clinical presentation in geriatric hypertension management. This scenario was designed to represent a typical patient in geriatric practice, where multiple comorbidities and polypharmacy are the norm rather than the exception.

Expert panel composition: Three specialist physicians participated in the study. The selection criteria include: (i) board certification in their respective specialty, (ii) more than 10 years of clinical experience in hypertension management, (iii) active clinical practice at Damghan Hospital, and (iv) willingness to participate in the study and provide detailed assessments. The expert decision-making panel consisted of three domain specialists: DM1 (Internist), DM2 (Cardiologist), and DM3 (Urologist). The experts' characteristics are given in Table 1.

Table 1. Expert characteristics

Expert	Speciality	Years of Experience	Institution	Role in Hypertension Management
DM1	Internist	>15 years	Damghan Hospital	Primary care for hypertensive patients
DM2	Cardiologist	>15 years	Damghan Hospital	Specialist care for complex hypertension
DM3	Urologist	>12 years	Damghan Hospital	Hypertension in patients with urological conditions

Note: DM = Decision-maker.

Assessment procedure: Expert assessments were initially piloted in October 2024 using an eight-class questionnaire (Version 1.0, including the acute vasodilator sodium nitroprusside) to gather preliminary feedback on drug-class framing and criterion clarity. The pilot questionnaire required experts to rank from 1 (most preferred) to 8 (least preferred). Following the feedback, the drug classification was revised and a definitive elicitation round was conducted in November 2024 using a seven-class, tie-free questionnaire (Version 2.0), restricted to chronic oral therapies. The three experts independently ranked the seven drug classes (A_1 – A_7) against eight criteria (X_1 – X_8), assigning ranks from 1 (most preferred) to 7 (least preferred). Assessments were based on the clinical scenario, the experts' clinical experience and judgement, and clinical guidelines. All rankings and decision matrices used in the reported calculations were derived exclusively from the November 2024 round. Pilot responses were not included in any calculations or the Appendix; the definitive Version 2.0 matrices are provided in Appendix A2.

Experts were instructed to assign lower ranks to drug classes considered more favourable for each criterion. Ties were not permitted; each drug class was assigned a unique rank (1 to 7) for each criterion, and no responses were missing. The experts were also asked to confirm their understanding of the pharmacological distinctions between alpha-1 blockers and central alpha-2 agonists before giving their rankings.

Clinical vignette provided to experts: “Please consider a 72-year-old male patient with hypertension (Blood Pressure (BP) 152/88 mmHg on three measurements over two weeks), type 2 diabetes (HbA1c 7.2% on metformin

1000 mg twice daily), prior myocardial infarction (3 years ago, treated with percutaneous coronary intervention), and mild congestive heart failure (NYHA Class II, ejection fraction 45%). The patient has no known drug allergies, is currently taking metformin and aspirin, and has no significant renal impairment (eGFR 65 mL/min/1.73 m²). Please rank the following antihypertensive drug classes from 1 (most preferred) to 7 (least preferred) for initiating or adjusting therapy in this patient, considering the eight criteria listed. Please provide independent rankings based on your clinical experience and judgement.”

2.2 Criteria Definition and Operationalization

The eight evaluation criteria were selected based on a literature review and clinical input. Each criterion was operationally defined with clear directionality. For every criterion, a lower rank represents a more favourable outcome. All ranks were subsequently converted to benefit scores for TOPSIS. The criteria and their definitions are shown in Table 2.

Table 2. Evaluation criteria and definitions

Criterion	Symbol	Definition	Scale	Direction (1 = Best)
Doctor’s experience	X_1	Expert’s familiarity and confidence with the drug class	Ordinal (1–7)	Lower = more experienced/confident
Suitability for older patients	X_2	Clinical suitability for patients aged >60 years	Ordinal (1–7)	Lower = more suitable
Suitability for diabetes	X_3	Efficacy and safety in diabetic patients	Ordinal (1–7)	Lower = more suitable
Suitability for kidney disease	X_4	Efficacy and safety in CKD patients	Ordinal (1–7)	Lower = more suitable
Suitability for CHF	X_5	Efficacy and safety in CHF patients	Ordinal (1–7)	Lower = more suitable
Suitability for MI	X_6	Efficacy and safety in post-MI patients	Ordinal (1–7)	Lower = more suitable
Complication risk	X_7	Likelihood/severity of adverse effects	Ordinal (1–7)	Lower = lower risk
Rapidity of effect	X_8	Speed of BP-lowering onset	Ordinal (1–7)	Lower = faster onset

Note: CHF: Congestive Heart Failure; CKD: Chronic Kidney Disease; BP: Blood Pressure; MI: Myocardial Infarction.

Note on criterion X_2 : Although the hypothetical patient is older than 60 years, this criterion captures the relative suitability of different drug classes for use in older patients. For example, some agents (e.g., dihydropyridine CCBs) are particularly appropriate in older patients with isolated systolic hypertension, while others may carry higher risks of adverse effects (e.g., orthostatic hypotension with alpha-1 blockers). Thus, the criterion varies across alternatives, not across patients.

Note on criterion X_1 : “Doctor’s experience” is operationalized as the expert’s self-assessed familiarity and confidence with each drug class. Although inherently subjective, it is clinically relevant because prescribing patterns are influenced by physician comfort and experience. The criterion is not circular because experts were asked to rank the drug classes based on their overall clinical experience rather than on their evaluations of the decision framework.

2.3 Rank-Frequency Linear Assignment

The Borda count methodology, originally proposed by Jean-Charles de Borda in 1781 within the context of social choice theory, is a group decision-making technique for prioritizing a discrete set of alternatives when multiple experts express preferences across several evaluative dimensions [29–31]. A salient feature of the Borda approach is its ability to accommodate ordinal rankings directly, without imposing restrictive assumptions regarding scale uniformity or interval comparability [29, 31]. The axiomatic foundations of ordinal social choice functions and positional voting rules are detailed extensively by Moulin [29].

To establish a consensus preference structure from the ordinal rankings, the expert assessments were aggregated using rank-frequency linear assignment. The group agreement matrix $C = [c_{ik}]$ contains the total criterion weight accumulated whenever alternative i is assigned rank k , summed across all criteria and decision-makers. Thus, each entry is a weighted rank-position frequency, not a classical Borda score. For example, if an alternative receives rank 1 under two criteria with weights w_a and w_b , their contribution to its rank-1 entry is $w_a + w_b$. The consensus ordering maximizes the sum of the selected agreement entries, subject to assigning each alternative exactly one rank and each rank exactly one alternative [32, 33].

The Borda method works by assigning points to each alternative based on its rank position in each expert's ranking. In the standard Borda count, for m alternatives, the top-ranked alternative receives $m - 1$ points, the second-ranked receives $m - 2$ points, and so on, with the last-ranked alternative receiving 0 points. The total Borda score for each alternative is the sum of points received from all experts or across all criteria. The alternatives are then ranked by their total Borda scores [29–31]. The present study uses a different aggregation rule: it maximizes rank-position agreement through linear assignment rather than ordering alternatives by their total Borda scores.

In our implementation, criterion-weighted rank-position frequencies were used in the linear assignment model. The procedural steps are described below.

Step 1—Construction of individual ranking matrices: For each participating decision-maker (internist, cardiologist, urologist), a distinct ranking matrix of dimensions $m \times n$ (7 classes $\times$ 8 criteria) was constructed. The rows correspond to the seven drug classes and the columns correspond to the eight evaluation criteria. Each entry in this matrix reflects the ordinal position assigned to a specific drug class by a given decision-maker under a particular criterion. For example, if the internist ranked beta-blockers as 1st for suitability for MI, the entry in the beta-blockers row and the suitability-for-MI column would be 1.

Step 2—Formation of the assignment matrix: A square assignment matrix of dimensions $m \times m$ (7×7) was subsequently derived, in which the rows denote the alternative drug classes and the columns denote the rank positions (1 through 7). The elements of this matrix are populated by summing the criterion weights for which a particular drug class secured a specified rank. For instance, if beta-blockers were ranked 1st for two criteria with weights of 0.125 each, the element in the beta-blockers row and column 1 would be 0.250.

In this application, criterion weights were uniform ($w_j = 0.125$ for all eight criteria, per the verified entropy calculation in Section 3.1), and all three decision-makers were weighted equally (no differential expert weighting was applied). No ties occurred, since each expert was required to assign a unique rank from 1 to 7 for every criterion (Section 2.1). Each element c_{ik} was therefore computed as the sum of the criterion weight (0.125) over every (expert, criterion) pair—3 experts $\times$ 8 criteria = 24 pairs per alternative—in which alternative i was assigned rank k . The resulting 7×7 assignment matrix, computed directly from the raw ranking matrices shown in section A2 in the Appendix.

Solving the linear assignment problem in Eq. (1) against this matrix gives an optimal objective value of 7.875. Exhaustive evaluation of the $7! = 5,040$ feasible assignments identifies two equally optimal rankings. They differ only in the positions of beta-blockers and alpha-1 blockers, which exchange ranks 1 and 6; both solutions are reported in Section 3.2.

Step 3—Linear assignment optimization: The optimal orderings were obtained by maximizing total rank-position agreement under the assignment constraints. Eq. (1) gives the mathematical formulation. The linear assignment approach uses the rank-frequency matrix directly and does not require the score normalization used in TOPSIS [31–33].

A primary rationale for adopting rank-frequency linear assignment is its capacity to aggregate strictly ordinal rankings without asserting interval comparability or utility scale equivalence across decision-makers [29, 31]. While general social choice theory discusses Borda voting in political contexts [29, 30], its specific application here addresses multi-attribute evaluation where individual criteria span disparate qualitative and quantitative domains [31].

Rather than combining raw heterogeneous units (e.g., onset time in hours versus qualitative complication risk) into a unified utility function, each specialist transforms criteria evaluations into a complete preference ordering over the alternatives. The linear assignment formulation then solves for the consensus permutation that maximizes agreement with these individual preference orderings, thereby bypassing direct numerical scale integration [32, 33].

$$\begin{aligned}
 \max \quad & \sum_{i=1}^m \sum_{k=1}^m c_{ik} x_{ik} \\
 \text{s.t.} \quad & \sum_{k=1}^m x_{ik} = 1, \quad \forall i, \\
 & \sum_{i=1}^m x_{ik} = 1, \quad \forall k, \\
 & x_{ik} \in \{0, 1\}, \quad \forall i, k.
 \end{aligned} \tag{1}$$

Here, “s.t.” denotes “subject to”; c_{ik} is the total criterion weight for which alternative i received rank k , and x_{ik} is a binary variable indicating whether alternative i is assigned rank k . The first constraint assigns each alternative exactly one rank, and the second assigns each rank to exactly one alternative. There are $m = 7$ alternatives, $n = 8$ criteria and $L = 3$ decision-makers.

2.4 Entropy Method for Criterion Weighting

Shannon's information entropy was applied to quantify score dispersion within each expert's evaluation matrix and derive criterion weights through a fixed mathematical rule. Positional scoring rules aggregate individual preference rankings [29, 30], while linear assignment optimization provides a way of deriving a consensus ranking from individual preference matrices [32, 33].

The entropy method quantifies the degree of information diversity in each criterion from the distribution of performance scores across alternatives. The resulting weights remain dependent on the elicited inputs and the chosen score transformation.

The entropy weighting procedure follows these steps:

Step 1—Normalization: For each decision-maker, raw ranks r_{ij} were transformed into benefit scores $s_{ij} = 8 - r_{ij}$. The score matrix was then normalized to obtain p_{ij} :

$$p_{ij} = \frac{s_{ij}}{\sum_{i=1}^m s_{ij}}$$

This normalization transforms the raw scores into probability-like values that sum to 1 for each criterion.

Step 2—Entropy calculation: For each criterion j , the entropy value E_j is calculated as follows:

$$E_j = -\frac{1}{\ln(m)} \sum_{i=1}^m p_{ij} \ln(p_{ij})$$

The factor $1/\ln(m)$, with $m = 7$ alternatives, ensures that $0 \leq E_j \leq 1$. Entropy is high when scores are similar across alternatives and low when their distribution is concentrated.

Step 3—Degree of diversification: The degree of diversification d_j is calculated as follows:

$$d_j = 1 - E_j$$

The degree of diversification represents the amount of information contained in the criterion. A higher d_j indicates greater discriminatory power.

Step 4—Criterion weight: The weight w_j for criterion j is:

$$w_j = \frac{d_j}{\sum_{j=1}^n d_j}$$

The final weights are normalized to sum to 1.

For each expert, entropy weights were calculated from the transformed rank scores. The individual criterion weights were combined by the geometric mean and normalized to sum to 1. Because every criterion column is a permutation of the same seven ranks, this procedure produces equal criterion weights and does not quantify preference intensity or relative clinical importance.

2.5 Group Decision Aggregation

TOPSIS was initially introduced by Hwang and Yoon [33], operates on the fundamental principle that the most preferred alternative should simultaneously exhibit the shortest geometric distance from the positive-ideal solution (PIS) and the longest geometric distance from the negative-ideal solution (NIS). This method has been applied in healthcare research [34] and offers intuitive decision logic, computational tractability, and the ability to accommodate various measurement scales [35]. General methodological principles and cross-industry implementations are detailed in [20, 36–38].

TOPSIS requires numerical scores for normalization and distance calculation. In this application, the ordinal ranks were converted to scores using $s = 8 - r$ and treated as equally spaced numerical inputs. This conversion is a modelling assumption and does not imply that successive ranks represent equal differences in expert preference or clinical performance. The resulting distances and rankings should therefore be interpreted with this limitation in mind, as discussed in Section 4.3.

In our implementation, the TOPSIS algorithm incorporated the following sequential stages:

Normalization: For each decision-maker l , the transformed benefit scores were vector-normalized using Eq. (2). Here, i indexes the seven alternatives, j indexes the eight criteria, and l indexes the three experts.

$$n_{ij}^{(l)} = \frac{s_{ij}^{(l)}}{\sqrt{\sum_{i=1}^m (s_{ij}^{(l)})^2}} \quad (2)$$

Weighted normalization: Each normalized score was multiplied by the corresponding expert-specific criterion weight, as shown in Eq. (3).

$$\begin{aligned} v_{ij}^{(l)} &= w_j^{(l)} n_{ij}^{(l)} \\ W^{(l)} &= \text{diag}\left(w_1^{(l)}, \dots, w_n^{(l)}\right) \\ V^{(l)} &= N^{(l)} W^{(l)} \end{aligned} \quad (3)$$

Here, N is the 7×8 normalized matrix, W is the 8×8 diagonal matrix of criterion weights, and V is the 7×8 weighted normalized matrix, calculated separately for each expert.

Ideal solutions: All transformed scores are benefit-oriented, so the positive-ideal and negative-ideal values are the maximum and minimum weighted scores for each criterion within each expert's matrix, respectively:

$$\begin{aligned} v_j^{+(l)} &= \max_i v_{ij}^{(l)}, \\ v_j^{-(l)} &= \min_i v_{ij}^{(l)}, \end{aligned} \quad j = 1, \dots, n \quad (4)$$

Raw ranks run from 1 (best) to 7 (worst); the transformation $s = 8 - r$ makes higher scores more favourable. Ideal values and distances are calculated after this transformation and vector normalization.

Distance calculation: For each expert l , Euclidean distances from the positive-ideal and negative-ideal profiles were calculated across all $n = 8$ criteria using Eq. (5).

$$\begin{aligned} d_i^{+(l)} &= \sqrt{\sum_{j=1}^n \left(v_{ij}^{(l)} - v_j^{+(l)}\right)^2} \\ d_i^{-(l)} &= \sqrt{\sum_{j=1}^n \left(v_{ij}^{(l)} - v_j^{-(l)}\right)^2} \\ i &= 1, \dots, m; \quad l = 1, \dots, L \end{aligned} \quad (5)$$

Group aggregation: For each alternative, the three expert-specific positive-ideal distances and the three negative-ideal distances were separately aggregated using the geometric mean using Eq. (6), with $L = 3$.

$$\begin{aligned} D_i^+ &= \left(\prod_{l=1}^L d_i^{+(l)}\right)^{\frac{1}{L}} \\ D_i^- &= \left(\prod_{l=1}^L d_i^{-(l)}\right)^{\frac{1}{L}} \end{aligned} \quad (6)$$

The per-decision-maker procedure followed by geometric averaging of distances was used to obtain the primary TOPSIS results. A single-pass comparison, which averages the three experts' transformed score matrices before normalization and TOPSIS, is reported separately in Appendix A4.2 and Appendix A5. Both procedures produce the same ranking for this dataset, but their closeness coefficients differ by up to 0.022748, for diuretics (A_7). Agreement in ordering does not make the two procedures numerically equivalent.

Closeness coefficient: For each drug class, the group relative closeness coefficient was calculated using Eq. (7).

$$C_i = \frac{D_i^-}{D_i^+ + D_i^-} \quad (7)$$

Final ordering: Alternatives were ranked in descending order of the group closeness coefficient C_i .

All computational procedures were implemented using MATLAB version 7.0 (MathWorks Inc., Natick, MA) for matrix operations and General Algebraic Modeling System (GAMS) for solving the linear assignment model for rank-frequency aggregation. An executable Python implementation reproducing the supplied matrices and both TOPSIS procedures is provided in Appendix A5.

3 Results

The proposed framework was applied to the hypothetical clinical scenario described in Section 2.1. The eight evaluative criteria—physician experience, old age, diabetic status, prior kidney disease, prior congestive heart failure, prior myocardial infarction, complication of medication, and rapid effect—were incorporated into the decision matrix.

The three specialist physicians provided independent ordinal rankings for each drug class under each criterion. The symbols used for criteria and alternatives are given in Table 3.

Lower rank values (1–7) indicate more favourable outcomes for every criterion. For example, a drug ranked 1 for rapidity of effect has the fastest onset, while a drug ranked 7 has the slowest onset. After conversion using $s = 8 - r$, higher scores represent more favourable outcomes.

Table 3. Symbols for alternatives and criteria

Symbol	Alternative or Criterion
Symbols for alternatives	
A_1	Beta-blockers
A_2	Calcium channel blockers (CCBs)
A_3	Angiotensin II receptor blockers (ARBs)
A_4	Alpha-1 blockers
A_5	Angiotensin-converting enzyme (ACE) inhibitors
A_6	Central alpha-2 agonists
A_7	Diuretics
Symbols for criteria	
X_1	Doctor's experience
X_2	Suitability for older patients (age >60 years)
X_3	Suitability for diabetic patients
X_4	Suitability for patients with a history of kidney disease
X_5	Suitability for patients with a history of congestive heart failure
X_6	Suitability for patients with a history of myocardial infarction
X_7	Complication of medication (risk)
X_8	Rapidity of therapeutic effect

3.1 Entropy Weight Calculations

The entropy method was applied separately to the transformed rank-score matrix of each expert. The resulting criterion weights are presented in Table 4.

Table 4. Entropy weights for each decision-maker and group weights

Criterion	DM1	DM2	DM3	Group Weight (Geometric Mean, Normalised)
X_1 : Doctor's experience	0.125	0.125	0.125	0.125
X_2 : Suitability for older patients	0.125	0.125	0.125	0.125
X_3 : Suitability for diabetes	0.125	0.125	0.125	0.125
X_4 : Suitability for kidney disease	0.125	0.125	0.125	0.125
X_5 : Suitability for CHF	0.125	0.125	0.125	0.125
X_6 : Suitability for MI	0.125	0.125	0.125	0.125
X_7 : Complication of medication	0.125	0.125	0.125	0.125
X_8 : Rapidity of effect	0.125	0.125	0.125	0.125

Note: DM: Decision-maker; CHF: Congestive heart failure; MI: Myocardial infarction.

The group weight vector is the geometric mean of the three individual weight vectors, normalized to sum to 1: Group $W = (0.125, 0.125, 0.125, 0.125, 0.125, 0.125, 0.125, 0.125)$, $\Sigma = 1.000$.

All eight criteria receive equal weight because every expert's raw ranking matrix is a complete permutation of ranks 1–7 for each criterion; under these conditions the entropy method is mathematically guaranteed to yield identical entropy (and therefore identical weight) for every criterion.

All eight criteria received identical entropy weights (0.125). This is a direct mathematical consequence of the elicitation format and the per-expert weighting procedure: each expert supplied a complete, tie-free ranking of the seven drug classes for every criterion, so each transformed criterion column is a permutation of the same values. Shannon entropy is unaffected by permuting those values across alternatives.

Consequently, the entropy-weighting step cannot distinguish criteria under this procedure, and the resulting weights carry no information about relative criterion importance. This limitation is discussed further in Section 4.3.

3.2 Linear Assignment Rankings

Using equal criterion weights of 0.125 and the raw matrices in Appendix A2, the linear assignment model attained an optimal objective value of 7.875. Exhaustive evaluation of all 5,040 feasible assignments identified two equally optimal rankings, listed from highest to lowest assigned preference below:

Solution 1: Beta-blockers → ACE inhibitors → ARBs → CCBs → Diuretics → Alpha-1 blockers → Central alpha-2 agonists.

Solution 2: Alpha-1 blockers → ACE inhibitors → ARBs → CCBs → Diuretics → Beta-blockers → Central alpha-2 agonists.

The assignment model therefore does not identify a unique first-ranked class: beta-blockers and alpha-1 blockers exchange ranks 1 and 6 across the two equally optimal solutions. ACE inhibitors, ARBs, CCBs, diuretics and central alpha-2 agonists retain ranks 2, 3, 4, 5 and 7, respectively.

This non-uniqueness is a property of the supplied rankings and the rank-position agreement objective; selecting one solver output would conceal an equally optimal alternative. Equal criterion weighting means that no criterion receives a larger weight, but it does not remove this ambiguity. The position of alpha-1 blockers cannot be attributed specifically to BPH, which was absent from the vignette and for which no contemporaneous scoring rationale was recorded. Such an interpretation would be speculative.

3.3 Ranking

Using the entropy-derived weights for each specialist, the normalised and weighted matrices were constructed: W_1 (DM1) = W_2 (DM2) = W_3 (DM3) = (0.125, 0.125, 0.125, 0.125, 0.125, 0.125, 0.125, 0.125). Using these weights, the normalized and weighted decision matrices were computed. The PIS and NIS for each decision-maker, as well as the aggregated group separation distances, are summarized in Table 5 and Table 6.

Table 5. Results of PIS and NIS

DM ⁺ /DM [−]	X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	X ₇	X ₈
DM1 ⁺	0.074	0.074	0.074	0.074	0.074	0.074	0.074	0.074
DM1 [−]	0.011	0.011	0.011	0.011	0.011	0.011	0.011	0.011
DM2 ⁺	0.074	0.074	0.074	0.074	0.074	0.074	0.074	0.074
DM2 [−]	0.011	0.011	0.011	0.011	0.011	0.011	0.011	0.011
DM3 ⁺	0.074	0.074	0.074	0.074	0.074	0.074	0.074	0.074
DM3 [−]	0.011	0.011	0.011	0.011	0.011	0.011	0.011	0.011

Note: PIS: Positive-ideal solutions; NIS: Negative-ideal solutions; DM: Decision-maker.

Table 6. Separation distances of group decision making

Alternative	DM1		DM2		DM3		GDM	
	d_i^+	d_i^-	d_i^+	d_i^-	d_i^+	d_i^-	D_i^+	D_i^-
Beta-blockers (A_1)	0.14	0.09	0.12	0.10	0.12	0.10	0.13	0.10
CCBs (A_2)	0.07	0.12	0.07	0.13	0.06	0.13	0.07	0.12
ARBs (A_3)	0.05	0.14	0.04	0.15	0.04	0.15	0.05	0.15
Alpha-1 blockers (A_4)	0.11	0.11	0.12	0.07	0.11	0.10	0.11	0.09
ACE inhibitors (A_5)	0.08	0.12	0.06	0.14	0.08	0.12	0.07	0.13
Central alpha-2 agonists (A_6)	0.14	0.07	0.16	0.07	0.16	0.07	0.15	0.07
Diuretics (A_7)	0.13	0.07	0.13	0.07	0.13	0.06	0.13	0.07

Note: DM: Decision-maker; GDM: Group decision-making (geometric mean). CCBs: Calcium channel blockers; ARBs: Angiotensin II receptor blockers; ACE: Angiotensin-converting enzyme.

3.4 Closeness Coefficients and Final Ranking

The group closeness coefficients were calculated from the unrounded group distances using Eq. (7) and are shown in Table 7.

Ordering these values in descending sequence gave the following final TOPSIS ranking:

(1) ARBs → (2) CCBs → (3) ACE inhibitors → (4) Alpha-1 blockers → (5) Beta-blockers → (6) Diuretics → (7) Central alpha-2 agonists.

The TOPSIS ranking differs from the linear assignment results because the methods use different aggregation rules. Linear assignment maximizes weighted rank-position agreement, whereas TOPSIS computes distances using

transformed rank scores. ARBs rank first by TOPSIS and third in both optimal assignment solutions. Central alpha-2 agonists rank last under both approaches, but the assignment model does not identify a unique first-ranked class. This agreement at the bottom of the rankings is descriptive and does not establish clinical validity. The equal entropy weights and the non-unique assignment optimum further limit interpretation; the findings remain a methodological demonstration based on three experts' assessments rather than a clinically validated prioritization.

Table 7. Closeness coefficients

Alternative	D^+	D^-	C_i
Beta-blockers (A_1)	0.13	0.10	0.43
CCB (A_2)	0.07	0.12	0.65
ARBs (A_3)	0.05	0.15	0.76
Alpha-1 blockers (A_4)	0.11	0.09	0.45
ACE inhibitors (A_5)	0.07	0.13	0.64
Central alpha-2 agonists (A_6)	0.15	0.07	0.31
Diuretics (A_7)	0.13	0.07	0.34

Note: CCBs: Calcium channel blockers; ARBs: Angiotensin II receptor blockers; ACE: Angiotensin-converting enzyme.

4 Discussion

The primary objective of this study was to develop a structured decision-support framework for antihypertensive selection in an older patient with multiple comorbidities, using two group MCDM approaches applied to a hypothetical scenario. TOPSIS ranked ARBs first, whereas linear assignment produced two equally optimal rankings with either beta-blockers or alpha-1 blockers first; both approaches placed central alpha-2 agonists last.

4.1 Interpretation of the Rankings

The first-place TOPSIS ranking of ARBs and the two equally optimal assignment rankings should be interpreted as outputs of the specified model, rather than endorsements of clinical superiority. ARBs are guideline-supported first-line options, including for patients who cannot tolerate ACE inhibitors because of cough [14, 16]. Alpha-1 blockers are not recommended as first-line therapy for hypertension [16], yet they occupy either first or sixth position across the two equally optimal assignment solutions. This non-uniqueness prevents a single first-place conclusion from the assignment model.

Beta-blockers rank fifth by TOPSIS and either first or sixth under linear assignment. The 2023 ESH guidelines retain beta-blockers among the major drug classes suitable for initiating and continuing treatment, particularly when compelling indications are present [14], whereas the 2025 AHA/ACC guideline does not recommend them as first-line therapy for uncomplicated hypertension [16]. The vignette includes prior myocardial infarction and heart failure, making this distinction relevant to interpreting the results.

The 2024 ESC guidelines emphasize risk-based treatment and early combination therapy [15]. This guidance provides clinical context but does not validate the calculated ranks or closeness coefficients.

Central alpha-2 agonists (methyldopa, clonidine) ranked last under both approaches: seventh in both optimal linear assignment solutions and seventh by TOPSIS. The 2025 AHA/ACC guideline generally reserves these agents for later-line use or selected clinical circumstances, reflecting their adverse-effect profiles, including sedation, dry mouth, and the risk of rebound hypertension following abrupt withdrawal of clonidine [16].

Diuretics ranked in the middle-to-lower range under both approaches (fifth in both optimal linear assignment solutions and sixth by TOPSIS). These positions reflect the experts' rankings for the selected scenario and do not establish a change in prescribing trends or clinical effectiveness. Thiazide and thiazide-like diuretics remain first-line options for hypertension, either alone or in combination [15, 16]. Their established benefits must be considered alongside potential metabolic adverse effects, including hypokalemia and hyperglycemia, and individual tolerability. The present drug-class ranking cannot distinguish the different indications and effects of thiazide, loop, and potassium-sparing diuretics.

4.2 Integration With Emerging Digital Decision Support Systems

The structured MCDM framework presented in this study aligns seamlessly with the ongoing digital transformation of healthcare delivery. Clinical decision support systems that incorporate multiple patient-specific parameters—including electronic health record data, real-time laboratory results, and predictive risk scores—are gaining momentum as tools to enhance prescribing safety and effectiveness.

Beyond clinical practice guidelines, preclinical research examines mechanisms of renal injury and potential protective interventions. A preclinical meta-analysis reported protective effects of ACE inhibitors and ARBs in

animal models of renal ischemia-reperfusion injury, while emphasizing the need for further work before clinical translation [39].

Recent technological developments highlight the integration of digital health solutions and predictive analytics into routine cardiovascular care. In September 2025, Cleveland Clinic announced an AI-powered clinical platform designed to optimize hypertension management and streamline complex therapy titration, with prospective evaluations planned [40]. According to the institutional announcement, the solution was developed using data from more than 400,000 patients and 10.5 million encounters [41]. The planned prospective trial evaluating safety metrics and blood pressure control trajectories is registered under ClinicalTrials.gov (NCT07218198). The registry record, last updated on April 28, 2026 and accessed on September 21, 2026, listed the study as not yet recruiting, with primary completion estimated for July 1, 2027 [42].

In a 2023 study, Herzog et al. [43] reported an AI model for predicting individualized antihypertensive treatment success, with a precision of 51.7%, recall of 44.4%, and F1-score of 47.8%. Model validation and agreement with guidelines were assessed in a 1,000-patient cohort.

4.3 Limitations and Future Directions

Several limitations warrant explicit acknowledgment. First, the expert panel comprised only three specialists from a single academic institution, which may introduce contextual bias and limit generalizability. Larger, multi-centre panels would enhance robustness. Second, this study employed a decision-modelling approach based on a representative clinical scenario rather than actual patient data. This limits the direct clinical applicability of the findings. No patient cohort, inclusion criteria, or prevalence data are reported to support assertions about population characteristics.

Third, the analysis was restricted to the drug-class level and did not differentiate between individual drugs within the same class, overlooking potentially meaningful intra-class differences in efficacy, tolerability, and pharmacokinetics—differences that are increasingly recognized as clinically significant. Fourth, our analysis was confined to a single hypothetical 72-year-old patient, restricting generalizability to younger hypertensive populations with different risk profiles and treatment goals. Fifth, and most importantly, the entropy weighting approach as implemented here has a specific mathematical limitation: because entropy was calculated separately for each expert and each expert provided a complete, tie-free ranking (a permutation of ranks 1 to 7) for every criterion, Shannon entropy—which depends only on the set of values in a distribution, not their assignment to alternatives—is mathematically guaranteed to be identical across all criteria. This produced equal weights (0.125) for all eight criteria, meaning the entropy-weighting step contributed no discriminatory information in this study; the reported rankings are therefore effectively equal-weighted aggregations of the experts' ordinal judgments rather than importance-weighted ones. Future applications seeking differentiated criterion weights should examine alternative elicitation and weighting approaches that can capture differences in criterion information or importance. Cardinal performance measurements or rankings that permit ties may yield different distributions, but do not by themselves guarantee differentiated or clinically meaningful weights. The linear assignment objective also has two equally optimal solutions that exchange the first and sixth positions of beta-blockers and alpha-1 blockers. A single returned solver solution therefore does not establish a unique consensus ranking. Beyond this structural issue, entropy weighting in general does not capture patient-reported preferences, quality-of-life considerations, or economic constraints. The transformation of ordinal ranks to cardinal values in TOPSIS should also be interpreted with appropriate caution.

Future research should address these limitations by including patient cohorts from diverse age groups, ethnicities, and care settings; incorporating pharmacogenomic data and machine learning algorithms to enable truly personalised therapy selection; and integrating dynamic patient-reported outcomes and cost-effectiveness estimates into the decision framework. Additionally, prospective validation of the model in a randomised controlled trial or pragmatic clinical implementation study would provide evidence for assessing its clinical utility. The incorporation of AI-driven approaches to automatically update criterion weights based on emerging clinical data and outcomes represents an exciting avenue for future development.

5 Conclusion

This study presents an exploratory group decision-support framework for prioritizing antihypertensive drug classes in a representative clinical scenario involving a 72-year-old patient with multiple comorbidities. The framework incorporates assessments from three clinicians, seven drug classes, and eight evaluation criteria, using entropy weighting alongside rank-frequency linear assignment and TOPSIS. Its methodological contribution lies in organizing expert judgments and clinical considerations within a common analytical structure, making the criteria and aggregation procedures explicit and providing a basis for examining how different decision-making approaches influence drug prioritization.

The application remains preliminary and specific to the selected scenario and expert panel. Any resulting rankings should be interpreted in relation to the input assessments, weighting procedure, and aggregation method, rather than

as evidence of the clinical superiority of individual drug classes. The small expert panel, reliance on ordinal assessments, and absence of patient-level outcomes or external validation limit the generalizability and clinical interpretation of the findings. The study therefore does not establish treatment efficacy, safety, or improvements in prescribing outcomes. All eight entropy weights are equal in this elicitation design, and the two equally optimal assignment solutions prevent a unique first-place conclusion from that model.

Further work should involve larger and more diverse expert panels, sensitivity analyses of the weighting and aggregation procedures, and evaluation across additional clinical scenarios. Differences between individual drugs within each class and patient preferences also warrant consideration. External validation and prospective clinical evaluation are needed to determine whether the framework can provide useful support for individualized treatment decisions alongside clinical guidelines and professional judgment.

Author Contributions

Conceptualization, H.F.; methodology, Z.H.; software, Z.H.; validation, H.F.; formal analysis, Z.H.; investigation, H.F.; data curation, H.F.; writing—original draft preparation, Z.H.; writing—review and editing, H.F.; visualization, H.F. All authors have read and agreed to the published version of the manuscript.

Informed Consent Statement

Written informed consent was obtained from all participating physicians for data collection and publication of their anonymized expert assessments.

Ethical Approval

Formal ethics committee approval was not required under institutional guidelines. An exemption determination for this expert-opinion study was documented by the Institutional Review Board of Damghan University (Protocol Ref: DU-IE-2024-09). The study involved anonymized expert assessments of a hypothetical clinical scenario, with no patient data collection, medical record review, patient contact, or clinical interventions.

Data Availability

The data supporting our research results are included within the article.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

The authors declare that no generative AI or AI-assisted technologies were used in the preparation of this manuscript.

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Appendix

Expert Elicitation Instruments, Raw Matrices, Data Preprocessing, and Computational Code

A1 Survey Instruments & Clinical Vignette

A1.1 Standardized Clinical Scenario (Vignette)

Target Patient Profile:

Age/Demographics: 72-year-old male.

Primary Diagnosis: Hypertension (blood pressure 152/88 mmHg on three measurements over two weeks).

Comorbidities: Type 2 diabetes mellitus (HbA1c 7.2% on metformin 1000 mg twice daily), history of prior myocardial infarction (3 years prior, treated with percutaneous coronary intervention), mild congestive heart failure (NYHA Class II, ejection fraction 45%), and no significant renal impairment (eGFR 65 mL/min/1.73 m²). The patient has no known drug allergies and is currently taking metformin and aspirin.

Task for Expert Evaluator: Rank seven major antihypertensive drug classes (A_1 – A_7) across eight clinical and operational criteria (X_1 – X_8).

A1.2 Questionnaire Version 1.0 (Round 1: October 12–20, 2024)

Note: Questionnaire v1.0 included acute emergency agents (e.g., sodium nitroprusside) alongside chronic oral therapies. Following expert panel feedback, these agents were removed to focus exclusively on chronic outpatient management.

Instructions (v1.0)

Please assign a rank from 1 (Most Preferred / Best) to 8 (Least Preferred / Worst) for each drug class under each evaluation criterion based on your clinical expertise.

Because Version 1.0 served only as a pilot instrument and its responses were not used in any reported calculation, its raw scores are not included in the Appendix; only the definitive Version 2.0 (Round 2) matrices, which underlie all reported results, are provided in Appendix A2.

Table A1. Drug classes in questionnaire version 1.0

Code	Drug Class/Alternative
A_1	Beta-blockers
A_2	Calcium channel blockers (CCBs)
A_3	Angiotensin receptor blockers (ARBs)
A_4	Alpha-1 blockers
A_5	ACE inhibitors
A_6	Central alpha-2 agonists
A_7	Diuretics
A_8	Direct acute vasodilators (sodium nitroprusside)

A1.3 Questionnaire version 2.0 (round 2: November 5–15, 2024)

The final instrument was used for the primary calculations. Acute emergency agents were excluded.

The final questionnaire included seven chronic oral antihypertensive drug classes (A_1 – A_7), excluding acute emergency agents. The eight evaluation criteria are defined in Table 2, and the symbols for the alternatives and criteria are listed in Table 3.

All three experts independently assessed the same standardized clinical vignette described in Section 2.1 and Appendix A1.1. For each criterion, they assigned each alternative a unique rank from 1 (most preferred) to 7 (least preferred), with no ties permitted.

A2 Raw Expert Evaluation Matrices

Tables A2–A4 present the raw ordinal preference rankings provided by DM1 (internist), DM2 (cardiologist), and DM3 (urologist), respectively. These matrices were obtained using questionnaire version 2.0 and constitute the input data for all reported calculations.

Table A2. Raw ordinal preference rankings provided by DM1 (internist)

Alternative	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
A_1 : Beta-blockers	7	6	6	7	1	1	7	5
A_2 : CCB	3	2	4	3	4	5	2	2
A_3 : ARBs	2	3	2	1	3	3	1	4
A_4 : Alpha-1 blockers	1	1	5	6	6	6	3	3
A_5 : ACE inhibitors	4	4	1	2	2	2	4	6
A_6 : Central alpha-2 agonists	5	5	7	5	7	7	6	1
A_7 : Diuretics	6	7	3	4	5	4	5	7

Note: DM: Decision-maker; CCBs: Calcium Channel Blockers; ARBs: Angiotensin II Receptor Blockers; ACE: Angiotensin-Converting Enzyme.

Table A3. Raw ordinal preference rankings provided by DM2 (cardiologist)

Alternative	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
A_1 : Beta-blockers	6	5	7	6	1	1	6	4
A_2 : CCB	2	1	3	4	5	4	2	2
A_3 : ARBs	1	3	2	1	3	3	1	3
A_4 : Alpha-1 blockers	4	4	5	5	6	6	4	5
A_5 : ACE inhibitors	3	2	1	2	2	2	3	6
A_6 : Central alpha-2 agonists	7	6	6	7	7	7	7	1
A_7 : Diuretics	5	7	4	3	4	5	5	7

Note: DM: Decision-maker; CCBs: Calcium Channel Blockers; ARBs: Angiotensin II Receptor Blockers; ACE: Angiotensin-Converting Enzyme.

Table A4. Raw ordinal preference rankings provided by DM3 (urologist)

Alternative	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
A_1 : Beta-blockers	6	6	6	6	1	1	6	4
A_2 : CCB	2	2	3	3	4	4	3	2
A_3 : ARBs	1	3	2	1	3	3	1	3
A_4 : Alpha-1 blockers	3	1	5	5	6	6	2	5
A_5 : ACE inhibitors	4	4	1	2	2	2	4	6
A_6 : Central alpha-2 agonists	7	5	7	7	7	7	7	1
A_7 : Diuretics	5	7	4	4	5	5	5	7

Note: DM: Decision-maker; CCBs: Calcium Channel Blockers; ARBs: Angiotensin II Receptor Blockers; ACE: Angiotensin-Converting Enzyme.

The weighted rank-frequency assignment matrix derived from these rankings is presented in Table A5. It accounts for all 24 expert–criterion pairs per alternative, and each row and column sums to 3.0 (24×0.125). The maximum total assigned weight is 7.875, attained by the two optimal orderings reported in Section 3.2.

Table A5. Weighted rank-frequency assignment matrix used in the linear assignment optimization

Alternative	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5	Rank 6	Rank 7
Beta-blockers	0.750	0.000	0.000	0.250	0.250	1.250	0.500
CCB	0.125	1.125	0.750	0.750	0.250	0.000	0.000
ARBs	1.000	0.500	1.375	0.125	0.000	0.000	0.000
Alpha-1 blockers	0.375	0.125	0.375	0.375	0.875	0.875	0.000
ACE inhibitors	0.375	1.250	0.250	0.750	0.000	0.375	0.000
Central alpha-2 agonists	0.375	0.000	0.000	0.000	0.500	0.375	1.750
Diuretics	0.000	0.000	0.250	0.750	1.125	0.125	0.750

Note: CCBs: Calcium Channel Blockers; ARBs: Angiotensin II Receptor Blockers; ACE: Angiotensin-Converting Enzyme.

A3 Preprocessing & Direction Conversion

To align raw rank inputs with TOPSIS benefit criteria maximization (where higher values indicate superior performance), raw ranks r_{ij} were transformed into benefit performance scores s_{ij} using:

$$s_{ij} = m + 1 - r_{ij} = 8 - r_{ij} \quad (\text{A1})$$

A4 Intermediate MCDM Tables

A4.1 Criterion entropy weights across decision-makers

Table A7 presents the entropy-derived criterion weights for the three decision-makers and the normalized group weights obtained by geometric aggregation.

A4.2 TOPSIS Intermediate Output Tables

For comparison, this section reports a single-pass TOPSIS procedure that averages the three experts' transformed score matrices before normalization. The primary procedure instead computes each expert's distances and then

Table A6. Transformed benefit score matrix (DM1)

Alternative	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
A_1	1	2	2	1	7	7	1	3
A_2	5	6	4	5	4	3	6	6
A_3	6	5	6	7	5	5	7	4
A_4	7	7	3	2	2	2	5	5
A_5	4	4	7	6	6	6	4	2
A_6	3	3	1	3	1	1	2	7
A_7	2	1	5	4	3	4	3	1

Note: DM: Decision-maker.

Table A7. Criterion entropy weights across decision-makers

Criterion	DM1 Weight (W_1)	DM2 Weight (W_2)	DM3 Weight (W_3)	Group Weight
X_1 : Doctor's experience	0.125	0.125	0.125	0.125
X_2 : Suitability for older patients	0.125	0.125	0.125	0.125
X_3 : Suitability for diabetes	0.125	0.125	0.125	0.125
X_4 : Suitability for kidney disease	0.125	0.125	0.125	0.125
X_5 : Suitability for CHF	0.125	0.125	0.125	0.125
X_6 : Suitability for MI	0.125	0.125	0.125	0.125
X_7 : Complication of medication	0.125	0.125	0.125	0.125
X_8 : Rapidity of effect	0.125	0.125	0.125	0.125
Sum	1.000	1.000	1.000	1.000

Note: DM: Decision-maker; CHF: Congestive heart failure; MI: Myocardial infarction.

combines distances by geometric mean (Section 2.5; Table 5, Table 6, and Table 7). Both procedures produce the same ranking in this dataset, but their closeness coefficients differ by up to 0.022748. Both procedures are implemented separately in Appendix A5.

Table A8. Vector-normalized aggregated matrix (R)

Alternative	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
A_1	0.143	0.200	0.141	0.142	0.593	0.593	0.142	0.311
A_2	0.485	0.542	0.396	0.397	0.310	0.310	0.482	0.510
A_3	0.571	0.428	0.509	0.595	0.423	0.423	0.595	0.396
A_4	0.457	0.514	0.254	0.227	0.169	0.169	0.425	0.311
A_5	0.371	0.400	0.593	0.510	0.508	0.508	0.368	0.170
A_6	0.143	0.228	0.113	0.142	0.085	0.085	0.113	0.594
A_7	0.228	0.086	0.367	0.369	0.282	0.282	0.255	0.085

Table A9. Weighted normalized matrix ($V = R \cdot W_{\text{Group}}$)

Alternative	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
A_1	0.018	0.025	0.018	0.018	0.074	0.074	0.018	0.039
A_2	0.061	0.068	0.049	0.050	0.039	0.039	0.060	0.064
A_3	0.071	0.054	0.064	0.074	0.053	0.053	0.074	0.050
A_4	0.057	0.064	0.032	0.028	0.021	0.021	0.053	0.039
A_5	0.046	0.050	0.074	0.064	0.063	0.063	0.046	0.021
A_6	0.018	0.029	0.014	0.018	0.011	0.011	0.014	0.074
A_7	0.029	0.011	0.046	0.046	0.035	0.035	0.032	0.011
PIS (V^+)	0.071	0.068	0.074	0.074	0.074	0.074	0.074	0.074
NIS (V^-)	0.018	0.011	0.014	0.018	0.011	0.011	0.014	0.011

Note: PIS: Positive-ideal solution (V^+); NIS: Negative-ideal solution (V^-); R : Vector-normalized aggregated decision matrix; W_{Group} : Group criterion-weight vector; V : Weighted normalized decision matrix.

A5 Executable Python Implementation Script

The following Python script uses NumPy and the Python standard library to reproduce the entropy weights, weighted rank-frequency assignment matrix, all optimal assignments, and the primary TOPSIS calculation from the raw matrices in Appendix A2. It also reproduces the single-pass TOPSIS comparison and intermediate matrices in Appendix A4.2. Rankings are calculated from unrounded values; numerical output is rounded only for display.

```
import itertools
import numpy as np

# Rows: A1-A7; columns: X1-X8. Raw ranks: 1 = best, 7 = worst.
DM1_raw = np.array([
    [7, 6, 6, 7, 1, 1, 7, 5],
    [3, 2, 4, 3, 4, 5, 2, 2],
    [2, 3, 2, 1, 3, 3, 1, 4],
    [1, 1, 5, 6, 6, 6, 3, 3],
    [4, 4, 1, 2, 2, 2, 4, 6],
    [5, 5, 7, 5, 7, 7, 6, 1],
    [6, 7, 3, 4, 5, 4, 5, 7],
])
DM2_raw = np.array([
    [6, 5, 7, 6, 1, 1, 6, 4],
    [2, 1, 3, 4, 5, 4, 2, 2],
    [1, 3, 2, 1, 3, 3, 1, 3],
    [4, 4, 5, 5, 6, 6, 4, 5],
    [3, 2, 1, 2, 2, 2, 3, 6],
    [7, 6, 6, 7, 7, 7, 7, 1],
    [5, 7, 4, 3, 4, 5, 5, 7],
])
DM3_raw = np.array([
    [6, 6, 6, 6, 1, 1, 6, 4],
    [2, 2, 3, 3, 4, 4, 3, 2],
    [1, 3, 2, 1, 3, 3, 1, 3],
    [3, 1, 5, 5, 6, 6, 2, 5],
    [4, 4, 1, 2, 2, 2, 4, 6],
    [7, 5, 7, 7, 7, 7, 7, 1],
    [5, 7, 4, 4, 5, 5, 5, 7],
])
raw = np.stack([DM1_raw, DM2_raw, DM3_raw])
assert np.all(np.sort(raw, axis=1) == np.arange(1, 8)[None, :, None])
scores = 8 - raw
labels = ["Beta-blockers", "CCB", "ARBs", "Alpha-1 blockers",
         "ACE inhibitors", "Central alpha-2 agonists", "Diuretics"]
```

```

def entropy_weights(matrix):
    p = matrix / matrix.sum(axis=0)
    entropy = -(p * np.log(p)).sum(axis=0) / np.log(matrix.shape[0])
    diversity = 1 - entropy
    return diversity / diversity.sum()

weights = np.array([entropy_weights(matrix) for matrix in scores])
assert np.allclose(weights, 0.125, atol=1e-14, rtol=0)
# Equal weights are exact for these permutations. Remove floating-point noise.
weights[:] = 0.125
group_weights = np.prod(weights, axis=0) ** (1 / 3)
group_weights /= group_weights.sum()

def topsis(matrix, w):
    r = matrix / np.linalg.norm(matrix, axis=0)
    v = r * w
    pis, nis = v.max(axis=0), v.min(axis=0)
    dp = np.linalg.norm(v - pis, axis=1)
    dm = np.linalg.norm(v - nis, axis=1)
    return r, v, pis, nis, dp, dm, dm / (dp + dm)

# Main analysis: separate expert distances, then their geometric means.
expert = [topsis(s, w) for s, w in zip(scores, weights)]
group_dp = np.prod([e[4] for e in expert], axis=0) ** (1 / 3)
group_dm = np.prod([e[5] for e in expert], axis=0) ** (1 / 3)
group_ci = group_dm / (group_dp + group_dm)
# Appendix variant: average scores, then one TOPSIS calculation.
single = topsis(scores.mean(axis=0), group_weights)

# Rank-position assignment, not classical Borda point scoring.
# Integer counts prevent floating-point errors from concealing equal optima.
counts = np.zeros((7, 7), dtype=int)
for matrix in raw:
    for i in range(7):
        counts[i] += np.bincount(matrix[i] - 1, minlength=7)
c = counts / 8
assert np.all(counts.sum(axis=0) == 24)
assert np.all(counts.sum(axis=1) == 24)
best, solutions = -1, []
for assignment in itertools.permutations(range(7)):
    value = sum(counts[i, assignment[i]] for i in range(7))
    if value > best:
        best, solutions = value, [assignment]
    elif value == best:
        solutions.append(assignment)

def display_matrix(title, matrix):
    print("\n" + title)
    for row in np.atleast_2d(matrix):
        print(" ".join(f"{x:.6f}" for x in row))

def display_ranking(title, dp, dm, ci):
    print("\n" + title + ": rank, alternative, D+, D-, Ci")
    # Sort full-precision coefficients; round only for display.
    for rank, i in enumerate(np.argsort(-ci), 1):
        print(f"{rank}: A{i + 1} {labels[i]} "
              f"{dp[i]:.6f} {dm[i]:.6f} {ci[i]:.6f}")

```

```

display_matrix("Table 4 / A4.1: DM1, DM2, DM3, group weights",
               np.column_stack([*weights, group_weights]))
display_matrix("Assignment matrix: rows A1-A7, columns ranks 1-7", c)
print(f"\nAssignment objective = {best / 8:.3f}")
print(f"Number of equal optimal assignments = {len(solutions)}")
for assignment in solutions:
    order = np.argsort(assignment) + 1
    print(" > ".join(f"A{i}" for i in order))
table5 = np.stack([row for e in expert for row in (e[2], e[3])])
display_matrix("Table 5: DM1+/-, DM2+/-, DM3+/-", table5)
table6 = np.column_stack(
    [d for e in expert for d in (e[4], e[5])] + [group_dp, group_dm]
)
display_matrix("Table 6: DM1+/-, DM2+/-, DM3+/-, group+/-", table6)
display_matrix("Table 7: rows A1-A7; columns D+, D-, Ci",
               np.column_stack([group_dp, group_dm, group_ci]))
display_ranking("Primary TOPSIS", group_dp, group_dm, group_ci)
display_matrix("A3: transformed DM1 scores", scores[0])
display_matrix("A4.2: normalized mean scores", single[0])
display_matrix("A4.2: weighted normalized mean scores", single[1])
display_matrix("A4.2: positive and negative ideals",
               np.stack([single[2], single[3]]))
display_ranking("Single-pass TOPSIS", single[4], single[5], single[6])
print("\nMaximum absolute difference between TOPSIS Ci values:",
      f"{np.max(np.abs(group_ci - single[6])):.6f}")

```